10. Scenario: You are working on a data visualization project and need to create basic plots using Matplotlib. You have a dataset containing the monthly sales data for a company, including the month and corresponding sales values. Your task is to develop a Python program that generates line plots and bar plots to visualize the sales data.

Question:

1. How would you develop a Python program to create a line plot of the monthly sales data?

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

sales = [1000, 1500, 1200, 1800, 2000, 2500]

plt.plot(months, sales, marker='o')

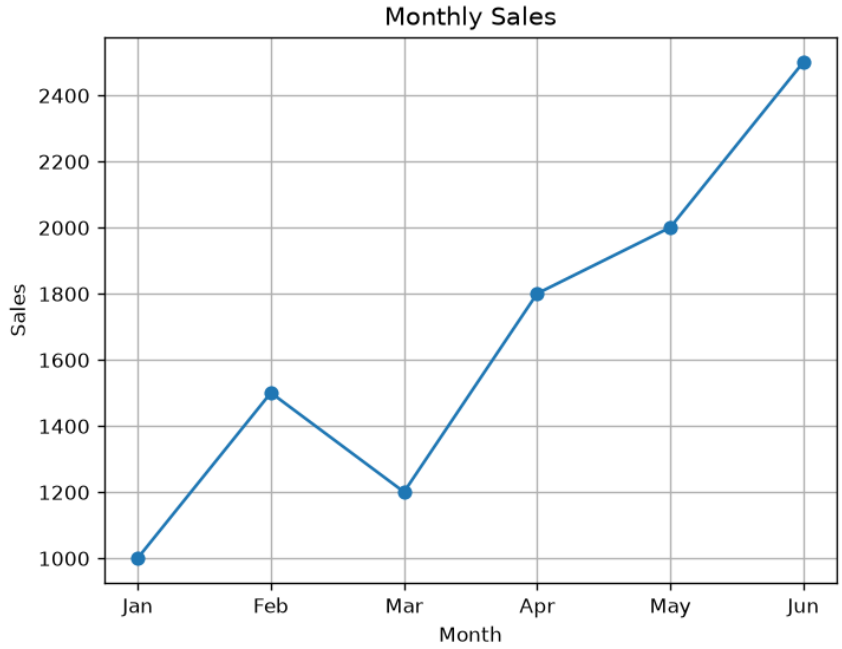
plt.title("Monthly Sales")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.grid(True)

plt.show()
```



2: How would you develop a Python program to create a bar plot of the monthly sales data?

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

sales = [1000, 1500, 1200, 1800, 2000, 2500]

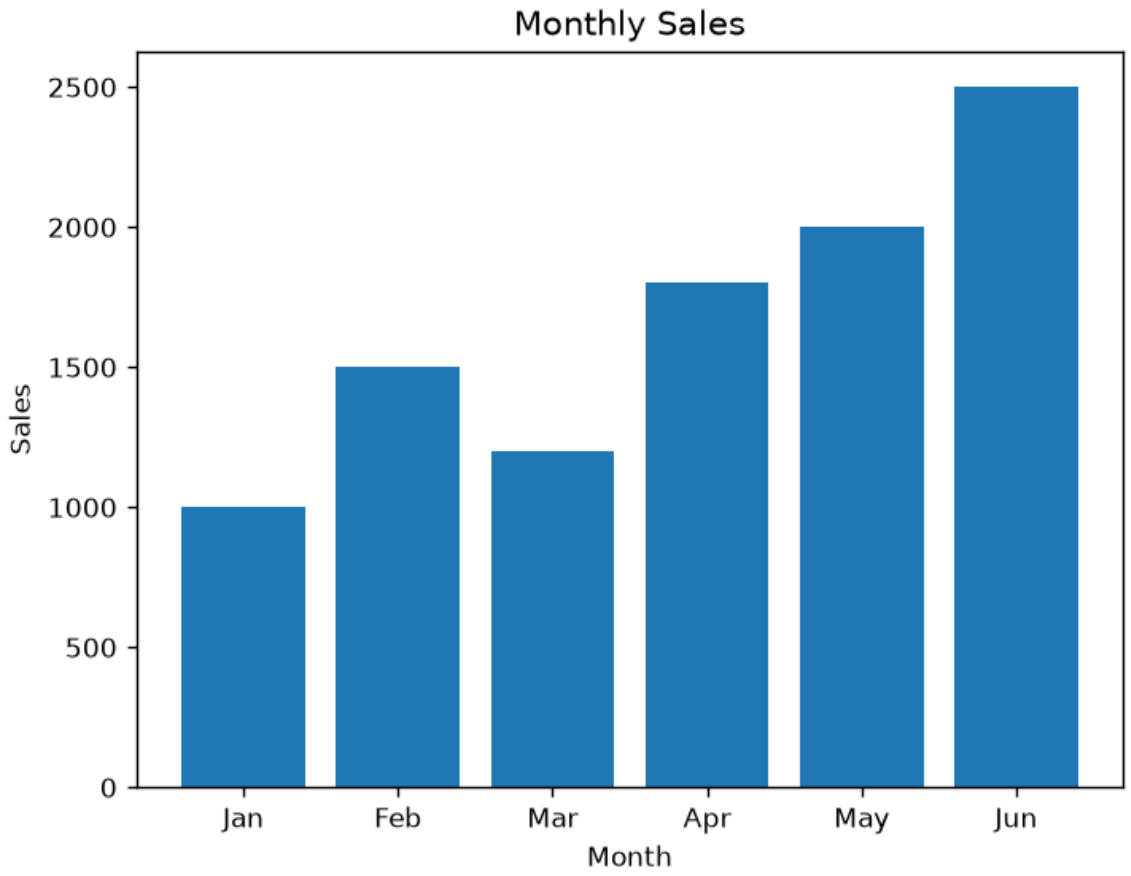
plt.bar(months, sales)

plt.title("Monthly Sales")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.show()
```



11. Scenario : You are a data scientist working for a company that sells products online. You have been tasked with creating a simple plot to show the sales of a product over time.

Question:

1. Write code to create a simple line plot in Python using Matplotlib to predict sales happened in a month?

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

sales = [100, 150, 130, 180, 200, 220]

plt.plot(months, sales, marker='o')

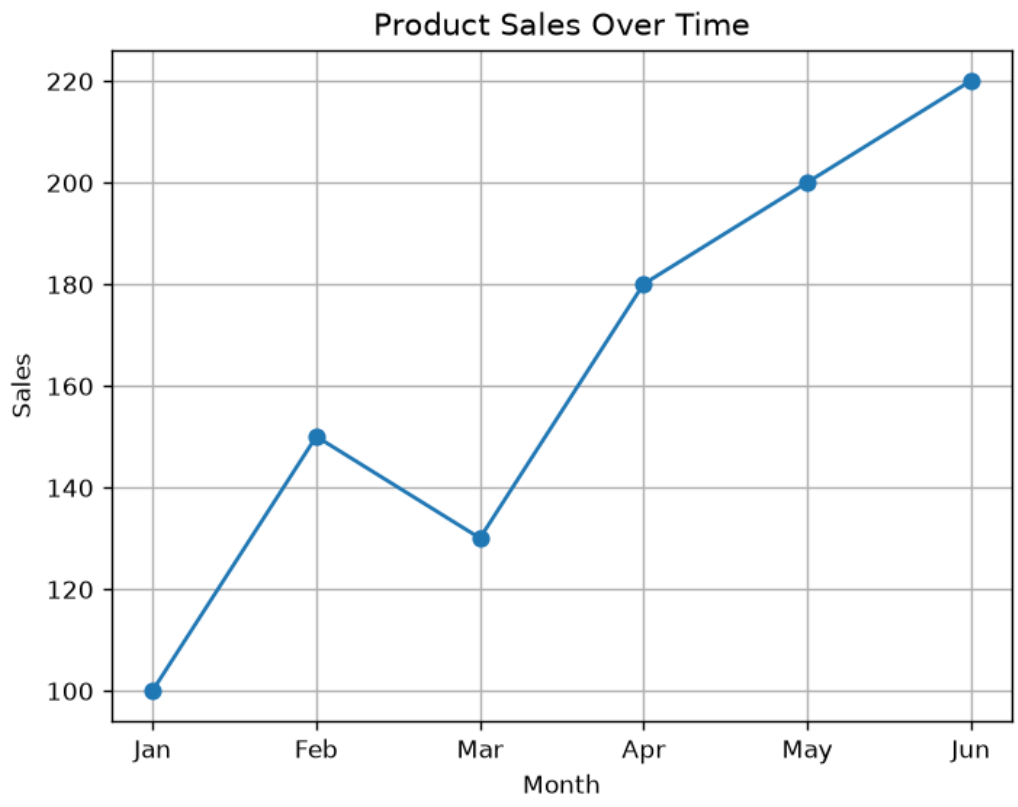
plt.title("Product Sales Over Time")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.grid(True)

plt.show()
```



2. Write code to create a scatter plot in Python using Matplotlib to predict sales happened in a month?

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

sales = [100, 150, 130, 180, 200, 220]

plt.scatter(months, sales)

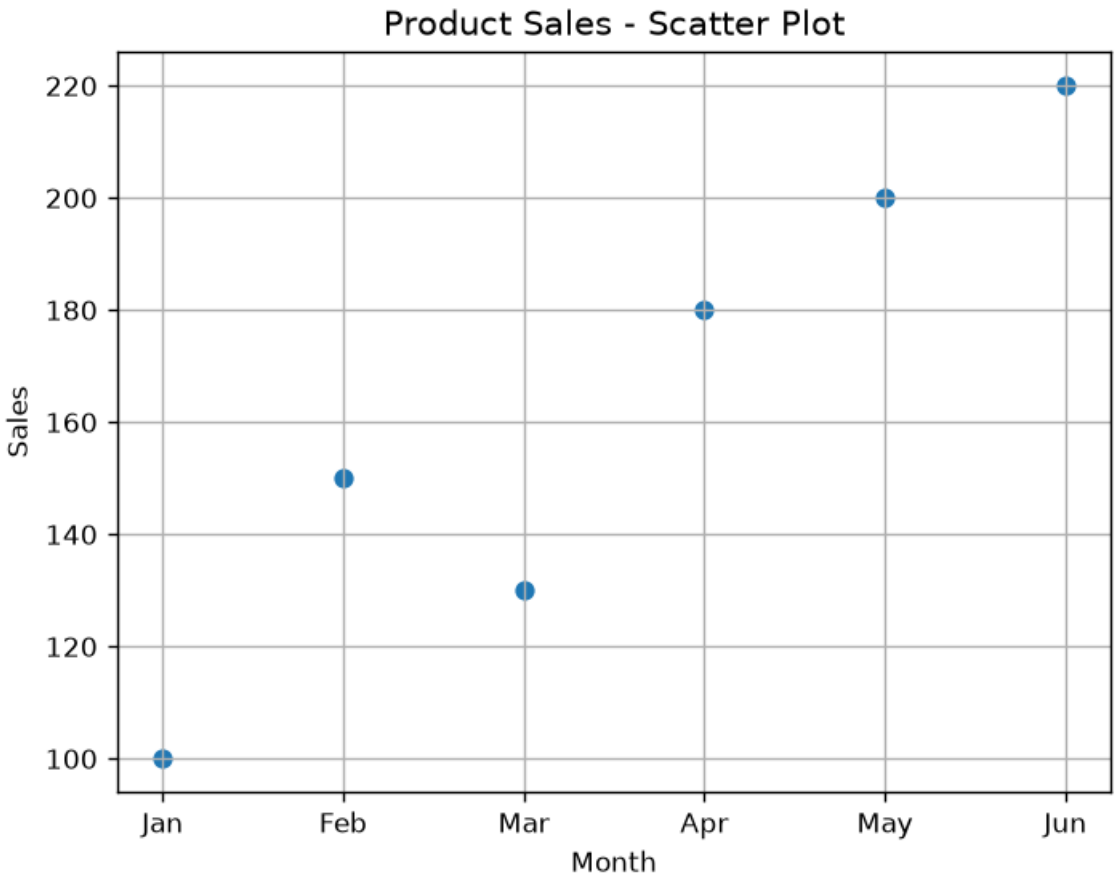
plt.title("Product Sales - Scatter Plot")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.grid(True)

plt.show()
```



3. Develop a Python program to create a bar plot of the monthly sales data.

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
```

sales = [100, 150, 130, 180, 200, 220]

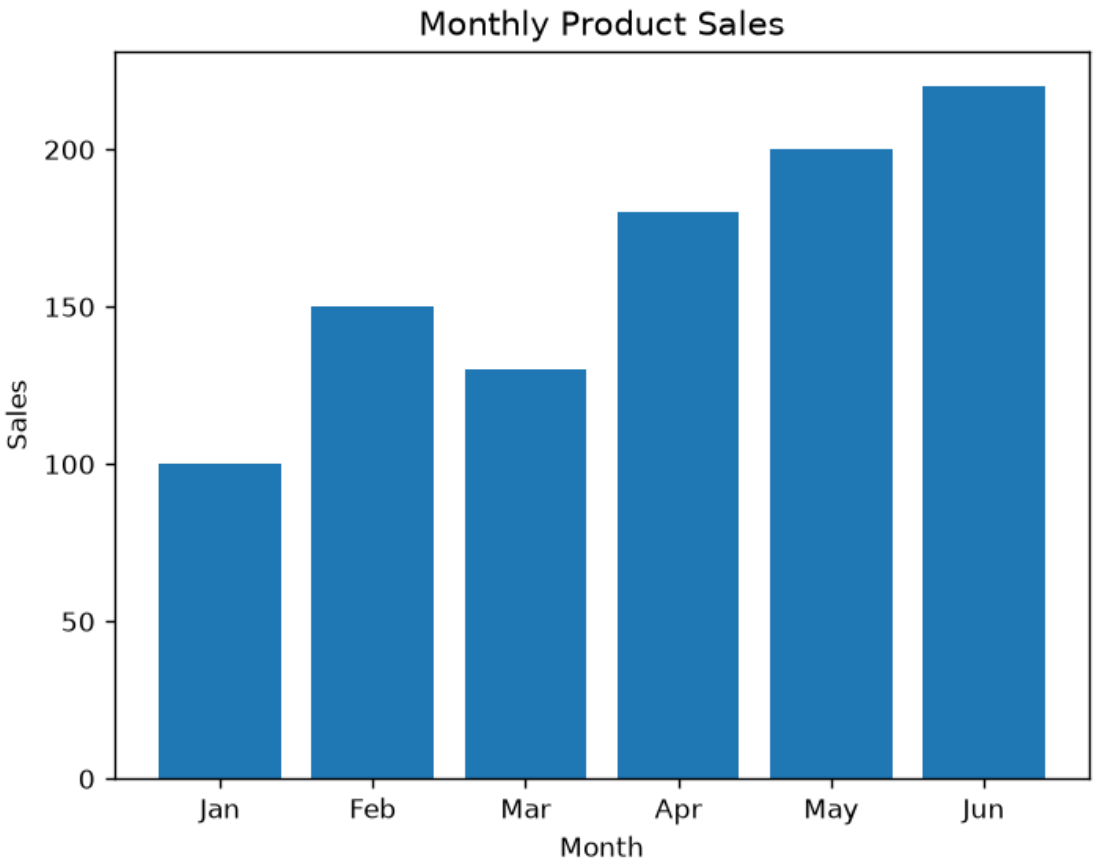
plt.bar(months, sales)

plt.title("Monthly Product Sales")

plt.xlabel("Month")

plt.ylabel("Sales")

plt.show()



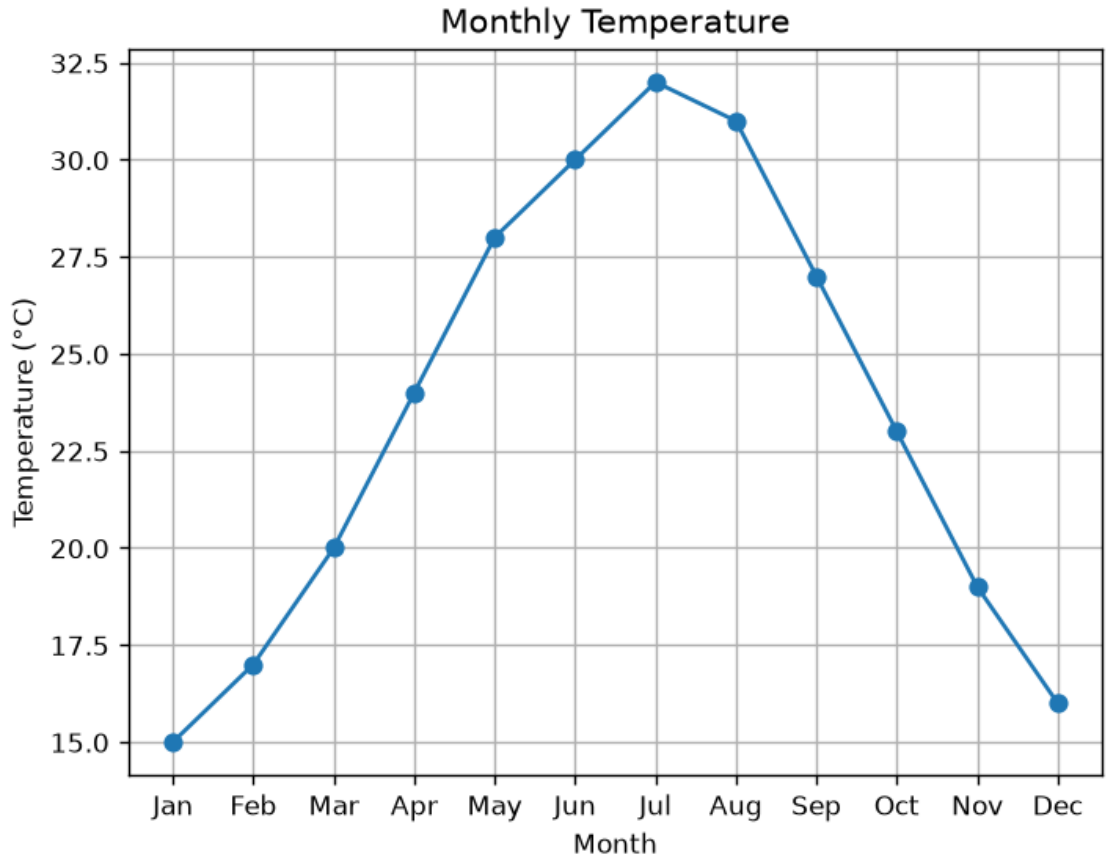
12. Scenario: You are working on a data analysis project that involves analyzing the monthly temperature and rainfall data for a city. You have a dataset containing the monthly temperature and rainfall values for each month of a year. Your task is to develop a Python program that generates line plots and scatter plots to visualize the temperature and rainfall data.

Question:

1. Develop a Python program to create a line plot of the monthly temperature data.

i**mport matplotlib.pyplot as plt**

```
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',  'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
temperature = [15, 17, 20, 24, 28, 30, 32, 31, 27, 23, 19, 16]
plt.plot(months, temperature, marker='o')
plt.title("Monthly Temperature")
plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
plt.grid(True)
plt.show()
```



2: Develop a Python program to create a scatter plot of the monthly rainfall data.

```
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',  'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
rainfall = [50, 40, 60, 120, 100, 90,70, 70, 60, 80, 70, 55]
```

```
plt.scatter(months, rainfall)
plt.title("Monthly Rainfall")
plt.xlabel("Month")
plt.ylabel("Rainfall (mm)")
plt.grid(True)
plt.show()
```

